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Original Study

Long-Term Care Needs Assessment: From Frameworks to Practice

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A B S T R A C T

Objectives: Identify and describe instruments used by countries to perform functional ability assessment as eligibility for their long-term care systems and compare them to existing healthy aging frameworks proposed by the World Health Organization (intrinsic capacity and functionality).
Design: Descriptive, case studies.
Setting and Participants: Country-level long-term care systems in 27 countries.
Methods: Selection of countries with long-term care systems in place and with expected increase in long-term care needs. For each country, data regarding the country features, as well as information of the long-term care systems, and the instruments to assess long-term care needs (functional ability) were collected. Features of the assessment tools were compared with the World Health Organization's intrinsic capacity and functionality domains, identifying the specific activities evaluated and the way the assessment is used to classify people and define benefits.
Results: Countries were classified into those with a long-term care system and a standardized instrument for assessing long-term care needs (group 1; n = 6), those that have long-term care initiatives do not have a standardized instrument to identify needs (group 2; n = 8), and those with no information on their systems or instruments for identify long-term care needs (group 3; n = 13). When looking at countries with standardized tools, instruments include several activities related to intrinsic capacity domains such as cognition, mobility, and psychological capacity; for functional ability, all domains except the ability to "contribute" are covered by the instruments.
Conclusions and Implications: Long-term care needs assessment is a complex but necessary task to be performed by long-term care systems. Countries embarking in the design and redesign of their systems can use the existing instruments, frameworks, and the experience from other countries to better adapt this process to their own needs and contexts.
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Population aging is a fact today. Among the several issues arising from this process, the increase in long-term care (LTC) needs is one of the most relevant challenges to be addressed. LTC is defined as activities undertaken by others to ensure that people with a significant ongoing loss of intrinsic capacity—the composite of all the physical and mental capacities that an individual can draw on—can maintain a level of functional ability consistent with their basic rights, fundamental freedoms, and human dignity.[1] According to the World Health Organization's (WHO's) healthy aging framework, functioning is determined not only by each individual's physical and mental capacities (intrinsic capacity) but also by the interactions of these capacities and the environment they inhabit.[1] Considering the problems associated with unmet needs related to LTC, WHO made a call for implementing an LTC system in every country in the world.[2]

However, designing and implementing an LTC system is not a simple task. One of the main issues required to move forward in implementing a system is the identification of people with LTC needs. Knowing the people with LTC needs is required not only to estimate the number of beneficiaries, services required, and costs, but it is also useful information to convince people and policymakers of the magnitude of the problem and evidence to advocate for the implementation of the system.[3] Any country embarked on the effort of implementing an LTC system needs to address the issue of assessing LTC needs in the population. Unfortunately, there is no consensus definition or measure.

LTC needs are related to concepts such as functional dependency, disability, and functional ability. Countries with LTC systems in place

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have developed their assessment tools; most of them are based on the idea of limitations to and/or need for help to perform activities of daily living (ADLs). However, the specific way in which these concepts are operationalized differs from case to case.

Considering the need of countries for implementing solutions to the increase in LTC needs, and the lack of a definition and tool to assess these needs, the aim of this document was to identify elements of LTC needs assessment tools currently used in countries around the world—which, in turn, requires identifying countries with LTC evaluations in place—as a way to identify common elements and differences between them, compare them with current LTC needs frameworks developed by WHO, and extract lessons that could help other countries in building their own LTC assessment instruments, understood as a necessary step into building their LTC systems.

Methods

Country Selection

The countries were selected using the following criteria.

1. Global representativeness: countries from the 6 WHO regions.
2. LTC system existence/LTC needs: within each region, countries were selected based on the following:
   a. Existence of a well-established LTC system in place. The rationale for this was maximizing the likelihood of finding information about LTC needs assessment instruments but also analyzing countries in which the evaluation of needs is carried out as part of an LTC system. An evident problem with this approach is the lack of a definition for LTC and LTC system. To overcome this obstacle, a first shortlist of 10 countries from 3 regions was selected from previous studies.[4,5]
   b. In those regions where no countries met the first criteria, the strategy was selecting countries with larger LTC needs. For this, we used the years lived with disability after aged 60, as a proxy for LTC needs in the country.[6] The rationale was that these countries would be more likely to have developed tools to deal with this issue (which in turn increases the probability of finding information on the instrument or LTC in general) or more interesting in adopting one in the near future.

Data Collection

For each one of the 27 selected countries, quantitative and qualitative data were collected. Information was grouped into 3 broad groups:

1. *Country features:* provides general context for each country to interpret results and make comparisons, including country name, WHO region, total population, share of population 60+, share of population 65+, and gross domestic product per capita.
2. *LTC system features:* to characterize the LTC systems in each country. What constitutes a formal LTC is not well-defined but information on the institutional setting and funding were collected, including LTC legal bodies/national policy, other relevant documents/policies (eg, disability laws, older people support), LTC funding scheme, responsible institution, LTC expenditure as share of total health expenditure, and LTC expenditure as share of GDP.
3. *LTC needs assessment tool features:* this information was intended to capture the details about the LTC needs assessment process in each country, including its application and use of results. Information gathered included name of the tool, target population, description, outcome/grading structure, and WHO functionality domains,[1] as well as intrinsic capacity domains from the integrated care for older people (ICOPE) tool.[7] For each instrument, the list of specific activities evaluated is included. These activities will be used to match with the domains proposed by the WHO frameworks. Finally, the analysis also includes the systematization of the criteria used for classifying people (for example, to define different levels of benefits). This allows, for each case, to understand the instrument's information requirements and the way in which this information is used for decision making. When possible, data on the LTC assessment were collected from national government websites, understood as official and updated information. The details on the sources of information for each country are presented in Supplementary Table 1.[8-57]

Country Classification

After collecting the information detailed previously, countries were classified according to the level of information available regarding LTC needs assessment. Grouping countries based on their LTC systems is tricky: because there is no consensus definition on what a LTC system is, any attempt to split countries into those with and without an LTC system can be subject to criticism. For this study, an LTC system is understood as a national system that ensures access to LTC services to the population that needs them.[3,58] Along with this, the existence of a particular legal body that regulates LTC services (at the national level) was also considered.

To overcome this problem, the strategy was using the available information to classify countries into those with and without systems but base the final classification into different groups on an objective measure: information on LTC needs assessment instruments. Using these criteria, the groups were as follows:

1. Group 1 - LTC needs assessment (standardized tool): countries with LTC systems in place identified, information of the LTC needs assessment process, and information on the specific features of the LTC needs assessment tool (national level). This gruop includes Germany, Japan, Republic of Korea, France, and Uruguay. Although the country does not have a national-level assessment tool, Finland was also included in this group, using the information of the "Care allowance for pensioners" (directly intended to cover people with "need of assistance" or "guidance and supervision" in activities of daily life).
2. Group 2 - LTC needs assessment but no standardized tool: countries with LTC initiatives in place, the existence of a LTC needs assessment tool but the instrument is not standardized at the national level (for example, several instruments or criteria within a country). This group includes the Netherlands, Denmark, Sweden, United Kingdom, United States, Canada, China, and South Africa.
3. Group 3 - No information on LTC system/assessment: no information on LTC initiatives at the national level. Countries in this group are India, Indonesia, Bangladesh, Thailand, Nigeria, Ethiopia, Algeria, Tanzania, Sudan, Egypt, Iran, and Morocco.

As shown in Figure 1, countries in group 3 do not have an LTC system in place and, consequently, no LTC needs assessment tool is available. On the other hand, group 1 countries are those with a clearly defined assessment tool, whether they are a country with a formal LTC system or with LTC-related initiatives provided through a different mechanism.[5,59-61] Finally, countries in group 2 include those in which a national-level standardized tool is not available. All these countries have some mechanism for assessing LTC needs but the tool is

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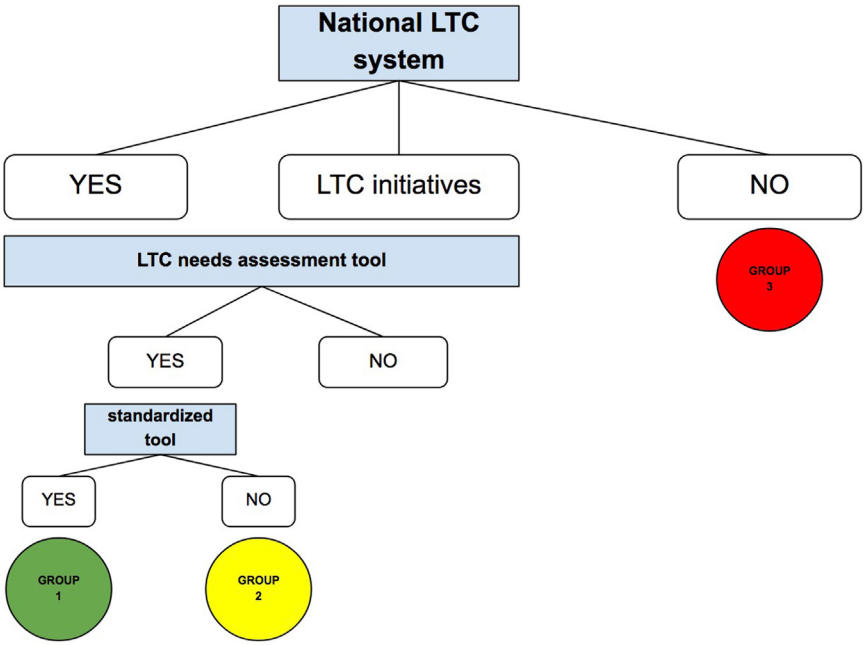


Fig. 1. Decision criteria for classifying countries into information groups.

- nonstandardized (individual case-to-case evaluation)
- carried out at the local level (meaning there is no one but several different instruments)
- different for different LTC-related benefits

Countries in this group are more heterogeneous in terms of their LTC institutionality. It includes countries with a well-defined LTC system and a national-level tool (Netherlands), countries with well-defined LTC services with national coverage but local-level assessment (Denmark, Sweden, United Kingdom, United States), countries with local LTC systems (Canada, China), and countries with several LTC-related services and no national-level coverage (South Africa).

The distribution of countries by WHO group and groups of information is presented in Table 1. It is also worth noting that, of the selected countries, those with a national-level assessment tool (group 1) are, on average, older (larger share of people 60+) and are more prone to establish LTC systems; however, because this information is incomplete and subject to selection bias, this hypothesis cannot be tested.

Countries and information groups correspond to the list of 27 countries selected and do not necessarily represent the actual reality of each region; in fact, countries were selected based on their likelihood of finding information of LTC needs assessment, meaning that groups 1 and 2 are expected to be overrepresented in this sample. However, the data on the older population follow the pattern exhibited by the different regions in terms of their population profiles, with regions like Africa and Asia having the lower share of older people and Europe and North America exhibiting the largest percentage of the population older than 60 years.[62]

Results

Information Groups and LTC Features

When available, additional information regarding the details of the tool (eg, specific activities considered or algorithm for grading functional dependency) was included.

One interesting feature of countries in group 1 is the existence of a well-defined LTC system: all these countries have an LTC-specific legal body (except Finland where the LTC entitlements are guaranteed in the Constitution) and half of them had implemented social insurance as a financing mechanism: Germany: Long-term Care Act (1995), and

Table 1
Country Groups of Information and Average Share of the Population Older Than 60 years in Selected Countries, by WHO Regions

WHO Region	Group 1 LTC Needs Assessment Available (Standardized Tool)		Group 2 LTC Needs Assessment But No Standardized Tool		Group 3 No Information on LTC System/ Assessment	
	No. of Countries	% People 60+	No. of Countries	% People 60+	No. of Countries	% People 60+
AFRO	0	N/A	1	8.54	6	5.72
EMRO	0	N/A	0	N/A	3	10.12
EURO	3	28.15	4	25.77	0	N/A
PAHO	1	20.24	2	23.89	0	N/A
SEARO	0	N/A	0	N/A	4	11.84
WPRO	2	28.74	1	17.35	0	N/A
Total	6	27.03	8	22.09	13	8.62

AFRO, African Region; EMRO, Eastern Mediterranean Region; EURO, European Region; PAHO, Americas Region; SEARO, South-East Asia Region; WPRO, Western Pacific Region.
Source: Authors based on United Nations Population Prospects.[62]

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long-term care strengthening acts I (2015) and II (2017); Japan: Long-Term Care Insurance Act (1997); Republic of Korea: Long-Term Care Insurance Act (2008); France: Allocation personnalisée d'autonomie, allocation personalisée à l'autonomie ("Personal autonomy allowance") (2002); and Uruguay: Law No. 19353. Creates the National System of Care (SNIC) (2015).[11,13,32,63,64]

Countries in group 2 are more heterogeneous including one country with LTC insurance but no national-level standardized tool (Netherlands), and a group of countries with tax-based LTC benefits and local-level assessment instruments (Denmark, Sweden, United Kingdom, United States, Canada, China, and South Africa). In terms of their system's legal support, these countries have broader legal bodies that justify the existence of LTC services—except the Netherlands, which has a LTC-specific legal body that regulates its LTC system: the Long-Term Care Act (WLZ)—such as social services laws (Denmark, Sweden, United Kingdom, and South Africa), health care laws (United States, Canada), or older adult-specific regulations (China and South Africa).[65]

Finally, countries in group 3 are those for which there is no evidence of the existence of a country LTC system and, consequently, no information on LTC needs assessment tools is available.

As stated before, countries in group 2 includes mostly systems in which LTC is explicitly addressed but the assessment is carried out at the local level (Netherlands, Denmark, Sweden, United Kingdom, and Canada), as well as countries in which there are some LTC-related initiatives but a LTC system—based on the existence of a specific legal body to address LTC needs—is not clearly in place (United States, China, and South Africa). Among these countries, the Netherlands in the only one that explicitly defines the existence of an LTC system (Long-Term Care Act); Denmark, Sweden, and the United Kingdom have no clear LTC system in place but define legal obligations to people with LTC needs as well as a specific mechanism for selecting beneficiaries to these benefits. Canada has no national LTC system but regional sub-systems (like the Ontario LTC system). The United States provides LTC services through Medicare and Medicaid, with no specific legal body addressing LTC needs, and linking coverage to health needs; for example, Medicare does not cover long-term care services if long-term care is the only care needed.[66] China has no national-level LTC system but has started pilots in several regions that work as regional sub-systems, including the publication of national-level guidelines for implementing LTC insurance pilots in the country. Finally, South Africa has introduced several LTC-related regulations in the past years, but they work as isolated initiatives with limited coverage.

LTC Assessments and Their Relation to Intrinsic Capacity and Functionality Domains

The 6 countries with information on LTC needs assessment tools were analyzed in terms of whether they consider domains proposed in the World Report on Ageing and Health and the ICOPE Handbook.[1,67]

As shown in Table 2, most countries include at least 4 functionality domains into their assessment, including mostly basic (BADLs) and instrumental activities of daily living (IADLs). Domain 5 ("Contribute") is not considered in the selected countries when assessing LTC needs. This issue highlights the difficulties of using the WHO functionality domains, as they are not mutually exclusive, and some activities used in the functionality assessment could overlap between domains.

In terms of the ICOPE domains, Table 3 shows the ICOPE elements considered in each tool. Most countries include cognition and mobility as part of their assessment tools. Domains of intrinsic and functional capacity mirror each other. This result is expected because intrinsic capacity is part of the functional ability.[1,67] Not surprisingly, activities used by the different LTC assessment tools fit in both tables. For example, most activities classified as "Mobility" in the ICOPE framework enter into the "Move around" domain of functionality (eg, moving inside/outside the house). One way to understand this relationship between the domains in the context of LTC assessment is thinking about the intrinsic capacity domains as related to "causes" and functionality domains as related to "consequences." For example, in the Korean case, the item "short-term memory" can be seen as "caused" by cognitive decline in the ICOPE framework but classifying as "causing" issues in the functionality domain "Learn, grow, and make decisions." Similarly, the item "behavior that may be self-endangering or endanger others" in the German tool falls into "Psychological capacity" (intrinsic capacity) and "Build and maintain relationships" (functional ability). These relationships are not necessarily "bijective" (one-to-one); for example, the item "Housekeeping" used in the Uruguayan case is related to "Mobility" (intrinsic capacity) and can be linked to "Move around" and "Meet their own basic needs" in the functional ability domains.

Table 2
LTC Assessment Tools (Selected Countries) and WHO Functionality Domains Considered

Country	LTC Assessment Instrument Name	WHO Functionality Domains Included*					
		Move Around	Build and Maintain Relationships	Meet Their Own Basic Needs	Learn, Grow and Make Decisions	Contribute	Total
Germany	New Assessment Procedure/Degree of Need	x	x	x	x		4
Japan	LTC requirement certification and support requirement certification	x	x	x	x		4
Republic of Korea	Cognitive Activities Support Rating of "Long-Term Care"	x	x	x	x		4
Finland	Care allowance for pensioners	x	x	x			3
France	National assessment grid: AGGIR (Autonomie gerontologie groupes iso-ressources)	x	x	x	x		4
Uruguay	Dependency scale (Baremo de dependencia)	x	x	x	x		4

Source: World Health Organization.[1]

*Notes: Ability to meet basic needs: This ability includes older people being able to afford an adequate diet, clothing, suitable housing, and health care and LTC services. It also extends to having support to minimize the impact of economic shocks that may come with illness, disability, and losing a spouse or the means of livelihood. Ability to learn, grow, and make decisions: includes efforts to continue to learn and apply knowledge, engage in problem solving, continue personal development, and be able to make choices. Ability to be mobile: refers to movement in all its forms, whether powered by the body (with or without an assistive device) or a vehicle. Mobility includes getting up from a chair or moving from a bed to a chair, walking for leisure, exercising, completing daily tasks, driving a car, and using public transport. Abilities to build and maintain relationships: a broad range of relationships is important to older people, including their relationships with children and other family members, intimate relationships, and informal social relationships with friends, neighbors, colleagues and acquaintances, as well as more formal relationships with community-service providers. Ability to contribute: covers a myriad of contributions that older people make to their families and communities, such as assisting friends and neighbors, mentoring peers and younger people, and caring for family members and the wider community.

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Table 3
LTC Assessment Tools (Selected Countries) and ICOPE Domains Considered

Country	LTC Assessment Instrument Name	ICOPE Domains of Intrinsic Capacity Included*					
		Cognition	Mobility	Sensory	Vitality/ Nutrition	Psychological Capacity	Total
Germany	New Assessment Procedure/Degree of Need	x	x			x	3
Japan	LTC requirement certification and support requirement certification	x	x	x		x	4
Republic of Korea	Cognitive Activities Support Rating of "Long-Term Care"	x	x			x	3
Finland	Care allowance for pensioners	x	x				2
France	National assessment grid: AGGIR (Autonomie gerontologie groupes iso-ressources)	x	x			x	3
Uruguay	Dependency scale (Baremo de dependencia)	x	x			x	3

Source: World Health Organization.[67]

*Notes: Cognition: Cognitive decline presents as increasing forgetfulness, loss of attention, and reduced ability to solve problems. Although the exact cause is not known, cognitive decline can be related to the aging of the brain, to diseases (for example, cardiovascular diseases, such as hypertension and stroke, or Alzheimer's disease), or even environmental factors such as a lack of physical exercise, social isolation, and a low level of education. Mobility: Limited mobility is a person's bodily capacity to move from one place to another and is termed locomotor capacity; Vitality/Nutrition: Physiological factors that contribute to an individual's intrinsic capacity. These may include energy balance and metabolism. Malnutrition is a key reason for decreased vitality in older age; Sensory: Including visual capacity, a critical component of intrinsic capacity, enabling people to be mobile and to interact safely with their peers and the environment, and hearing capacity, a critical part of monitoring older people's intrinsic capacity at the community level; Psychological capacity: applies to older adults who have 2 or more simultaneous symptoms of depression most of or all the time for at least 2 weeks, but who do not meet the criteria for a diagnosis of major depression.

Common Elements Used in LTC Needs Assessment

Table 4 presents the specific activities used by each tool in the selected countries (group 1). Although there are some differences between countries, it is also evident that many activities are the same for several instruments.

The 6 instruments use a combination of BADLs and IADLs. In the case of BADLs, they include activities proposed in the Barthel and Katz indexes.[68,69] On the other hand, many of the IADLs follow the activities proposed by Lawton and Brody (see Supplementary Table 2 for a list of the activities used by each instrument).[70] Finally, many tools also include other broad sets of activities, related to social activities and participation. These are mostly included in the "Build and maintain relationships" and "Learn, grow, and make decisions" domains, and are usually classified as advanced activities of daily living (AADLs) related to high cognitive functions and social activities (Supplementary Table 2).[71,72]

Finally, Table 5 shows how the activities are assessed and combined into an outcome that defines the level of LTC needs for each country. Most countries use several levels to classify autonomy, the dependency or need for help, depending on the focus of their instruments.

LTC needs grading is used to select beneficiaries into the system, as well as to define the benefits that each person can receive from the system. Countries like Germany and the Republic of Korea have moved recently from a 3-level classification system to 5 levels, in order to extend the definition of LTC needs and include people usually excluded from the system, such as people with dementia.[10,17] This highlights the need to take a broad approach when assessing LTC needs. For example, Japan's system explicitly includes mental health as part of the evaluation, and Uruguay uses a separate questionnaire to assess people with cognitive problems.

Discussion

The document aims to understand the way in which countries face the challenge to assess LTC needs in practice. The exercise intends to serve as an example for other countries asking how this LTC assessment should be performed.

However, in the process, several issues arose. First, when selecting the countries to be analyzed, it was clear that LTC, as well as LTC systems, are concepts that are not easily understood or defined. Asking whether a country has an LTC system in place is not an easy question to answer. Knowing which countries have LTC systems in place and which countries do not, is necessary to launch a process of implementing LTC systems around the world. Although there is no way to clearly identify the existence of an LTC system, elements like the definition of people in need, the existence of an instrument to identify and classify beneficiaries, as well as LTC-specific laws can be used to identify the presence of a "system" instead of LTC-related initiatives in a country.

Second, LTC needs assessment tools are aligned with the concepts of intrinsic capacity and functional ability, proposed by WHO. This result is expected because these tools look to identify people's need for help in performing ADLs. Both concepts (intrinsic capacity and functional ability) are also closely related and, consequently, their domains can be seen as 2 sides of the same coin. When looking at the activities evaluated by the LTC assessment tools, they can be seen from a "cause perspective" (linked to intrinsic capacity) or from a "consequence perspective," (linked to functional ability). In any case, the tools analyzed include most but not all the proposed domains. Need assessment does not consider domains like "Nutrition" (intrinsic capacity) and "Contribute" (functional ability"). Both domains appear as too difficult to assess in practice or out of scope when trying to evaluate people's LTC needs. On the one hand, "Nutrition" could be too general as a domain for functional ability: it is important for intrinsic capacity but also for health in general. On the other hand, "Contribute" seems too specific for assessing LTC needs: using the WHO definition, even people with no functional loss could fail in "contributing."

Third, most countries use a set of BADLs, IADLs, and AADLs to operationalize their definitions of LTC needs. Traditional scales (Barthel, Katz, Lawton and Brody) can be a good first step in looking for activities to be included in the assessment. Countries should decide which activities should be included, considering the LTC system and LTC needs assessment goals, as well as the required changes to adapt these activities to understand how they measure or not to LTC needs today. Because many of the proposed instruments were developed more than 50 years ago, countries should consider changes to update, including new activities, to avoid potential gender bias in their assessment.[43,73,74]

Finally, results and conclusions should be interpreted with caution because of limitations of the study. First, as mentioned before, the selection of cases is not complete and, consequently, interpretation of statistical information, such as the distribution of countries into

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Table 4
Activities Used to Assess LTC Needs in Selected Countries

Country	WHO Functionality Domains			
	Move Around	Build and Maintain Relationships	Meet Their Own Basic Needs	Learn, Grow, and Make Decisions
Germany	Mobility over short distances and the ability to transfer the position of the body; participation in social and, in a very broad sense, cultural activities (including mobility outside of the domestic setting).	Behavior that may be self-endangering or endanger others or involve other problems, including psychological problems such as anxiety, panic attacks, or delusions; budgeting time, adherence to a rhythm of being awake and sleeping, spending available time sensibly (corresponding to needs) and maintaining social relationships; participation in social and, in a very broad sense, cultural activities (including mobility outside of the domestic setting).	Personal hygiene, ability to get dressed, eat, and drink along with functions related to bladder and bowel movements; activities aimed at coming to terms with the demands and stress resulting from an illness or therapy measures, such as taking medications, dressing wounds, handling physical aids and appliances, and engaging in extended therapies within and beyond the domestic setting.	Memory, perception, thought, judgment, communication; household activities and regulation of necessary business dealings for everyday life (use of services, interaction with public authorities, financial affairs).
Japan	Paralysis, walking, transfer, mobility, going outdoors.	Communication, persecution tendency, confabulation, rapid changes in mood, repeat same story, shouting, resistance to care, restless, eager to go out by oneself, collection mania, violence, severe memory loss, abnormal muttering, selfish behavior, endless talking, taking drugs, difficulty of adapting to group activity.	Washing, cutting nails, swallowing, eating, urination, defecation, oral hygiene, washing face, hairdressing, dressing, sleep problems, resistance to care, taking drugs, shopping, cooking.	Communication, difficulty returning home, severe memory loss, taking drugs, handling finances, daily decision making, shopping, cooking.
Republic of Korea	Transfer from bed to chair, moving out of room; motor disturbance in right arm, left arm, right leg, left leg; limitation of movement in joints in shoulder, elbow, wrist, hip, knee, ankle.	Feeling persecuted, visual or auditory hallucinations, depressed mood, reversal of day and night, resisting advice or care, restlessness, being lost, verbal or physical violence, trying to go out alone, destroying items, meaningless behavior, hiding money or items, dressing inappropriately, unsanitary behavior.	Dressing, washing face, brushing teeth, bathing, eating, turning in bed, sitting up in bed, transfer from bed to chair, moving out of room, toileting, bowel control, bladder control. Nursing care needs: Tracheostomy care, suction, oxygen therapy, sore care, tube feeding, pain control, urine catheter care, colostomy care, dialysis care.	Short-term memory, remembering dates, places, date of birth, understanding instructions, judgment, communication.
Finland	Exercise.	Communication.	Dressing, eating, medical treatment, or personal hygiene and other comparable activities Personal activities, essential household chores, and out-of-home transactions.	
France	Dressing, moving inside house; moving outside house; using transportation.	Coherence, free time activities.	Orientation, toileting, eating, bathing, cooking, housekeeping, medical treatment.	Orientation, communication, managing money, using transportation, shopping, medical treatment.
Uruguay	Change and keep the body position; moving inside the house; moving outside the house; housekeeping.	Social and community life; communication skills and mental autonomy.	Eat and drink; personal hygiene and toileting; washing; other personal hygiene activities; dressing; health (self-care); housekeeping.	Health (self-care) Social and communitary life.

Source: Based on information from Germany[10]; Japan[14]; Korea[16–18]; Finland[23]; France[31,33]; and Uruguay.[36]

groups, must be taken as a reference not as an estimation. Second, classification of countries, particularly those with LTC in place, is also controversial. As discussed in the Methods section, just like there is no easy way for selecting cases, classification into groups is also subject to controversy.

The focus of the study is understanding the way countries measure and use LTC needs assessment in practice, and the limitations were approached to present information on different instruments used in different contexts. With these caveats in mind, the analysis needs to be considered as an exercise that, despite its limitations, provides useful information on concrete and practical implementation issues, like the way to assess LTC needs, suggesting questions and answers for a topic underexplored in terms of country-level policies.

Conclusions and Implications

The article highlights the need for implementing LTC needs assessment at the national level, as a required step into the building of a national LTC system.

Designing and implementing an LTC system is a complex task. Defining and measuring needs is also challenging, but lessons can be extracted from the experiences of other countries in this topic.

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Table 5
LTC Needs Grading by Country

Country	Outcome/Grading Structure
Germany	Five-level scale based on the assessment of 6 modules that combined give a score between 0 and 100. People in levels B1 to B5 are classified as in need of LTC care: • No need for LTC (0 to 9 points) • Degree of Need B1 (10 to 29 points) • Degree of Need B2 (30 to 49 points) • Degree of Need B3 (50 to 69 points) • Degree of Need B4 (70 and more points) • Degree of Need B4 + special constellation of need: Degree of Need B5.
Japan	Seven care-level categories that are further divided into 2 types: • "Care levels" (1–5): are for increasingly disabled individuals in need of LTC to help with BADLs. • "Support required" levels (1 and 2): for individuals who can live independently but are afraid of being in need of care and require assistance for IADLs.
Republic of Korea	Five levels: • Level 1: Requires help in all aspects of daily life • Level 2: Requires help in most parts of daily life • Level 3: Requires help in part of daily life • Level 4: Requires some help for daily living because of the functional disability • Level 5: Patients with dementia
Finland	Three levels (payment), based on the need for assistance, guidance, and supervision: • Care allowance at the basic rate • Care allowance at the middle rate • Care allowance at the highest rate
France	Six levels of dependency (groups 1 to 4 are entitled to benefits from allocation personalisée à l'autonomie) • GIR 1: older people confined to bed or chair, with seriously impaired intellectual functions, essential continual presence of health care personnel. • GIR 2: 2 groups of older people -Those confined to bed or chair, with intellectual functions not totally affected, but needing assistance for most IADLs -Those with affected mental functions but who are still mostly able to move around • GIR 3: older people who have retained their intellectual faculties and part of their ability to move around but who require assistance several times a day for their physical autonomy. Most of them need help going to the toilet. • GIR 4: - Those who are unable to get up on their own, but once up can move around indoors. They sometimes need help with washing and dressing. Most of them can eat without assistance; - Those who have no problem in moving around but need help with bodily functions and meals • GIR 5: people who can move around inside their home without assistance and can eat and dress themselves alone. They require occasional help with washing, preparing meals, and doing housework • GIR 6: people who have not lost their autonomy for daily living activities.
Uruguay	Three levels of functional dependency: • Severe/total • Moderate • Mild

GIR, Groupes iso-ressources.
Source: Based on information from Germany[10]; Japan[14]; Korea,[16-18]; Finland[23]; France[31,33]; Uruguay.[36]

Although there is still no consensus definition and measurement tool to assess LTC needs, several elements can be used to build new ways to evaluate for needs. There is still a wide range of flexibility in terms of activities to be assessed and ways to grade LTC needs. Countries should try to adapt these instruments considering their LTC system goals.

Disclosure

The author declares no conflicts of interest.

Supplementary Data

Supplementary data related to this article can be found online at https://doi.org/10.1016/j.jamda.2024.105409.

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